

AP CALCULUS AB - UNIT 2

Guided Notes 2.2: Tangent Lines and the Derivative



Name: _____ Class: _____ Date: _____

Learning Objectives and Essential Question

Essential question: How can we represent a derivative at a point and a derivative function as limits of difference quotients?

- Represent a derivative at a point and a derivative function as limits of difference quotients.
- Translate among $f'(a)$, y' , $\frac{dy}{dx}$, and related notation.
- Use a derivative value to write the equation of a tangent line.

Key Knowledge, Vocabulary, and Notation

Derivative at a point

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}, \text{ provided the limit exists.}$$

Derivative function

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$

Equivalent form

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}.$$

Tangent line

$$y - f(a) = f'(a)(x - a).$$

Concept Development and Strategy

1. The base point must remain fixed in a derivative-at-a-point limit. In the h -form, the nearby input is $a+h$; in the $x \rightarrow a$ form, the denominator is $x-a$.
2. Notation communicates both the function and the variable of differentiation. For example, $\frac{dP}{dt}$ is the rate of change of P with respect to t .
3. A derivative value supplies only the tangent slope. To write the tangent line, also use the point $(a, f(a))$.

Worked Example: Definition at a point

Use the limit definition to find $f'(3)$ for $f(x) = x^2 - 4x$.

Answer and Reasoning

$$\frac{(3+h)^2 - 4(3+h) - (-3)}{h} = \frac{2h+h^2}{h} = 2+h, \text{ so } f'(3) = 2.$$

Worked Example: Recognizing a derivative

Interpret $\lim_{x \rightarrow 2} \frac{\ln x - \ln 2}{x - 2}$.

Answer and Reasoning

It is the derivative of $f(x) = \ln x$ at $x = 2$, so its value is $f'(2) = \frac{1}{2}$.

Worked Example: Tangent equation

Given $g(4) = -1$ and $g'(4) = 3.5$, write the tangent line.

Answer and Reasoning

$$y + 1 = 3.5(x - 4).$$

Guided Practice and Discussion

Directions

Show the mathematical setup, preserve exact values unless a decimal is requested, and justify existence or interpretation statements with the appropriate definition or theorem.

1. Write $f'(5)$ using the h -form of the derivative definition.

2. Write $g'(a)$ using a variable x approaching a .

3. Use the definition to find $f'(1)$ for $f(x) = x^2 + 3x$.

4. Use the definition to find $g'(4)$ for $g(x) = \sqrt{x}$.

5. For $f(x) = \frac{1}{x}$, use the definition to derive $f'(x)$ for $x \neq 0$.

6. Use the definition to find the derivative function of $f(x) = x^2 - 2x + 4$.

7. If $f'(a)$ exists, prove $\lim_{h \rightarrow 0} \frac{f(a+2h) - f(a)}{h} = 2f'(a)$.

AP Reasoning Check

Multiple-Choice Check

Choose the best answer. Explain why at least one distractor is incorrect during class discussion.

1. Which limit equals $f'(4)$?

(A) $\lim_{h \rightarrow 0} \frac{f(4+h)-f(h)}{4}$

(B) $\lim_{h \rightarrow 0} \frac{f(4+h)-f(4)}{h}$

(C) $\lim_{h \rightarrow 4} \frac{f(h)-f(4)}{h}$

(D) $\lim_{x \rightarrow 0} \frac{f(x)-f(4)}{x-4}$

2. $\lim_{x \rightarrow 3} \frac{x^4-81}{x-3}$ represents

(A) the value of x^4 at 3

(B) the derivative of x^4 at 3

(C) the average rate on $[0, 3]$

(D) the derivative of x^3 at 4

3. For $f(x) = e^x$, a calculator approximation of $\frac{f(1.001)-f(1)}{0.001}$ is closest to

(A) 1

(B) 2.718

(C) 7.389

(D) 0

Common Errors and AP Precision

- State the derivative notation, input value, sign, and units when interpreting a rate in context.
- Check differentiability before using a tangent slope, and keep the order of Product and Quotient Rule terms accurate.
- A complete AP justification names the relevant definition, condition, or theorem rather than reporting only a numerical result.

Exit Ticket

For $f(x) = x|x|$, use the definition to determine $f'(0)$ even though the formula changes at 0.

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